

Obstetric and gynecological nursing (Practice)

No	procedures
1	Measure fundus height.
2	Abdominal grips.
3	Routes to give magnesium sulfate.
4	Vaginal examination.
5	Perineal care.
6	Use of partograph.
7	Uterine massage.
8	Breast self-examination.
9	Pap smear.



Dedication & Acknowledgment

This work is dedicated to the distinguished experts who contributed their sincere efforts and valuable expertise to the development and review of the nursing curricula for Secondary Technical Nursing Schools, playing a pivotal role in refining its academic and practical content to the highest standards. With sincere gratitude and appreciation to:

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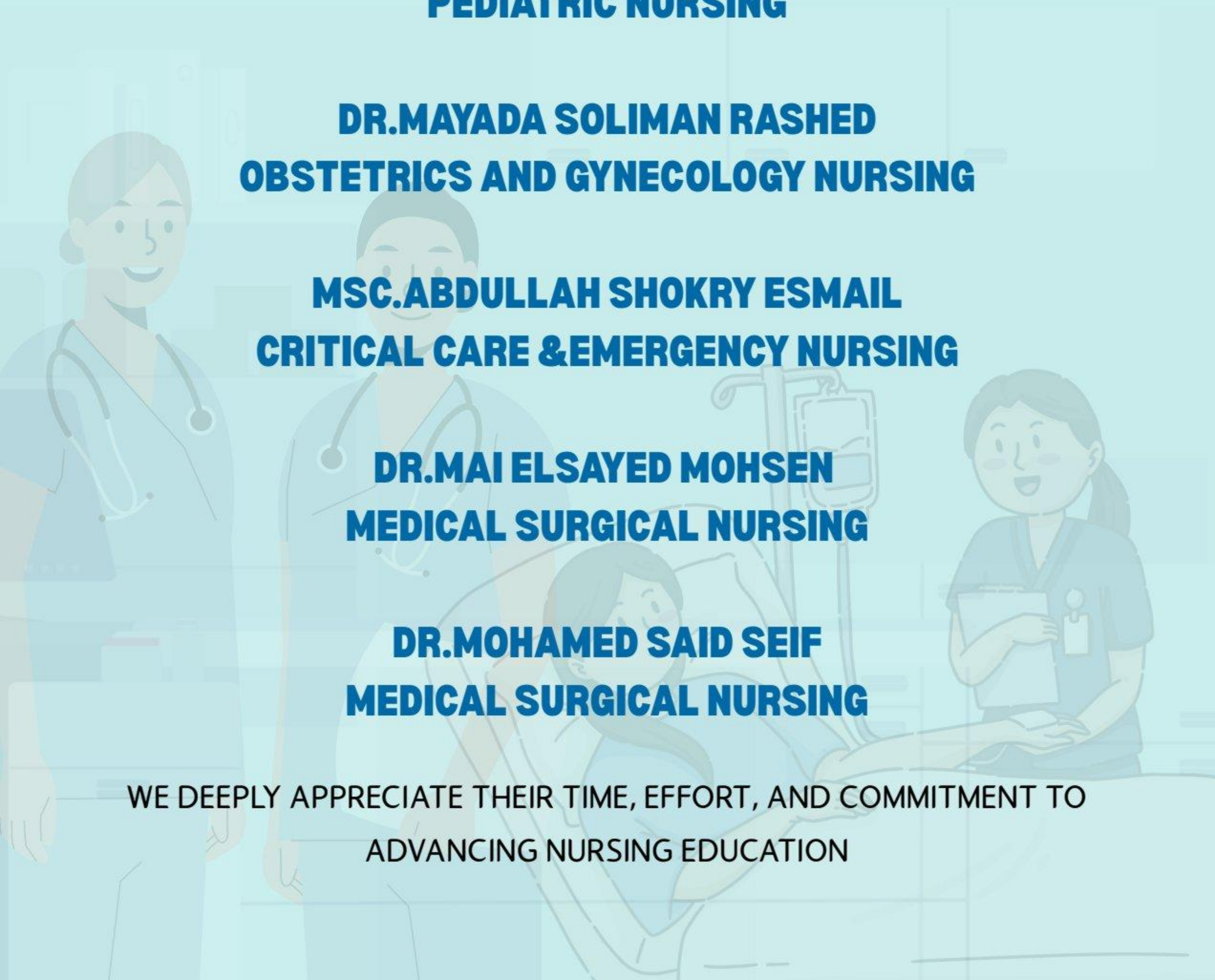
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WE DEEPLY APPRECIATE THEIR TIME, EFFORT, AND COMMITMENT TO
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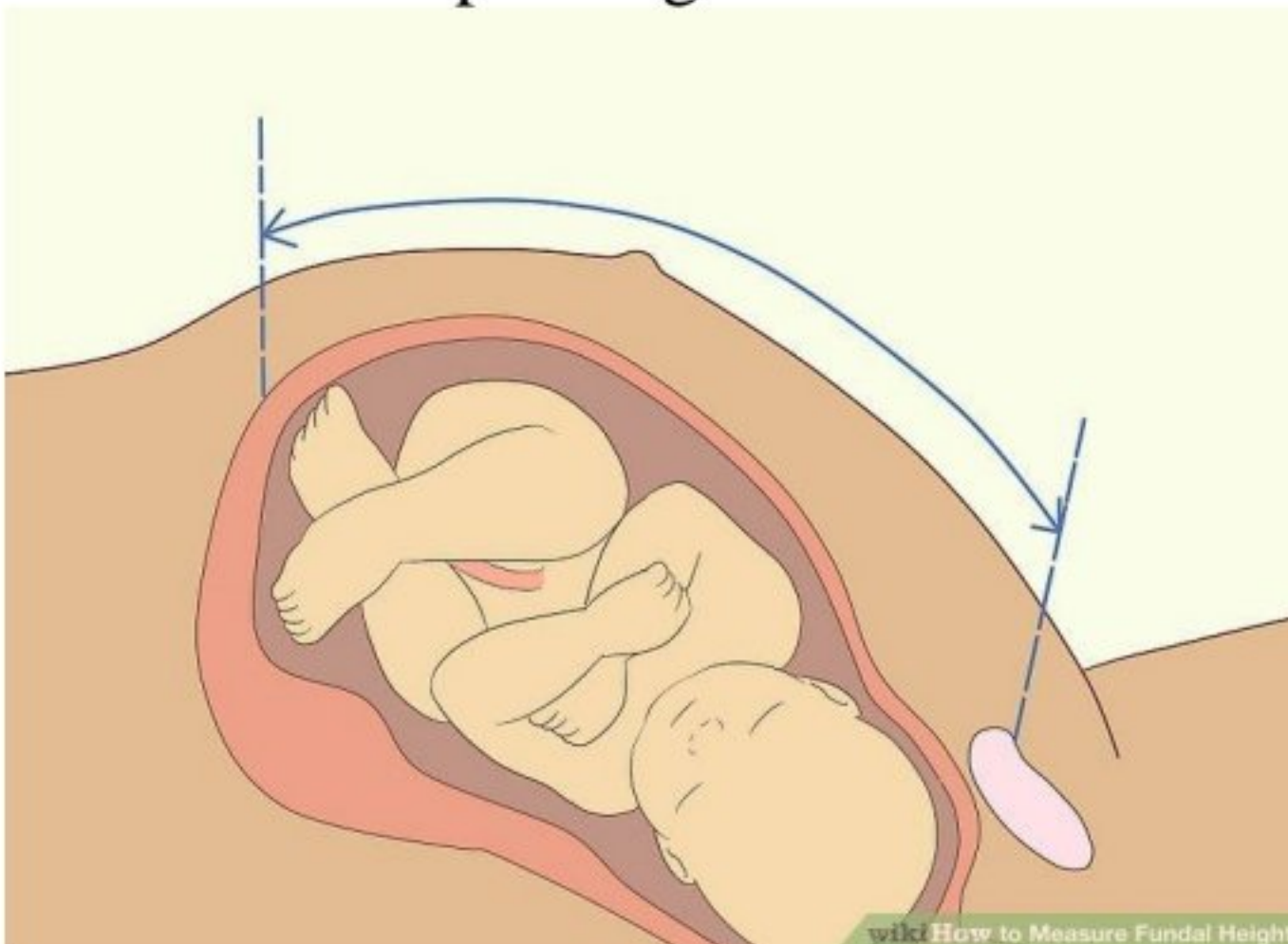
Measure fundus height.

Steps of procedure	Rational	
1- Ask the woman to completely empty her bladder immediately before proceeding with the abdominal examination. 	This is important as even a half full bladder might result in an increase in the fundal weigh	
2- Ask the woman to lie on her back with the upper part of her body supported by cushions. 	Never make a pregnant woman lie flat on her back as the heavy uterus may compress the main blood vessels returning to the heart and cause fainting (supine hypotension). Ask her to partially flex her hips and knees	
3- Stand on the right side of the woman to examine her in a systematic manner, The attention of the woman may be diverted by conversation.		
4- Your hand must be warm and should be placed on the abdomen till the uterus is relaxed before the palpation is actually begun.		
5- To measure the fundal height, place the ulnar (medial/inner) border of the hand on the woman's abdomen, parallel to the symphysis pubis. Start from the xiphisternum (the lower end of the sternum/breastbone), and gradually proceed downwards towards the symphysis pubis, lifting you hand between each step down, till your finally feel a bulge/resistance, which is the uterine fundus		



6-Mark the level of the fundus. Using a measuring tape (a tailor's tape measure which is made of non-stretchable material), measure the distance (in cm) from the upper border of the symphysis pubis to the top of the fundus. After 24 weeks of gestation, the fundal weight (in cm) corresponds to the gestational age in weeks (within 1-2 cm deviation). Remember, at the time of measuring the fundal height in cm, the legs of the woman should be kept straight and not flexed.

The supine position in late pregnancy and labor has also been shown to be associated with higher fundal height readings; therefore, this can give to false readings and an inaccurate estimate of the gestational age. It is therefore, recommended that the woman lies down in a half-lying position when measuring the fundal height



7-The normal fundal height is different at different weeks of pregnancy. To estimate the gestational age through the fundal height, the abdomen is divided into parts by imaginary lines. The most important one is the one passing through the umbilicus. Then divide the lower abdomen (below the umbilicus) into 3 parts with 2 equidistant lines between the symphysis pubis and the umbilicus. Similarly, divide the upper abdomen into three parts, again with two imaginary equidistant lines, between the umbilicus and the xiphisternum

Look where the fundal height is and judge as given below:

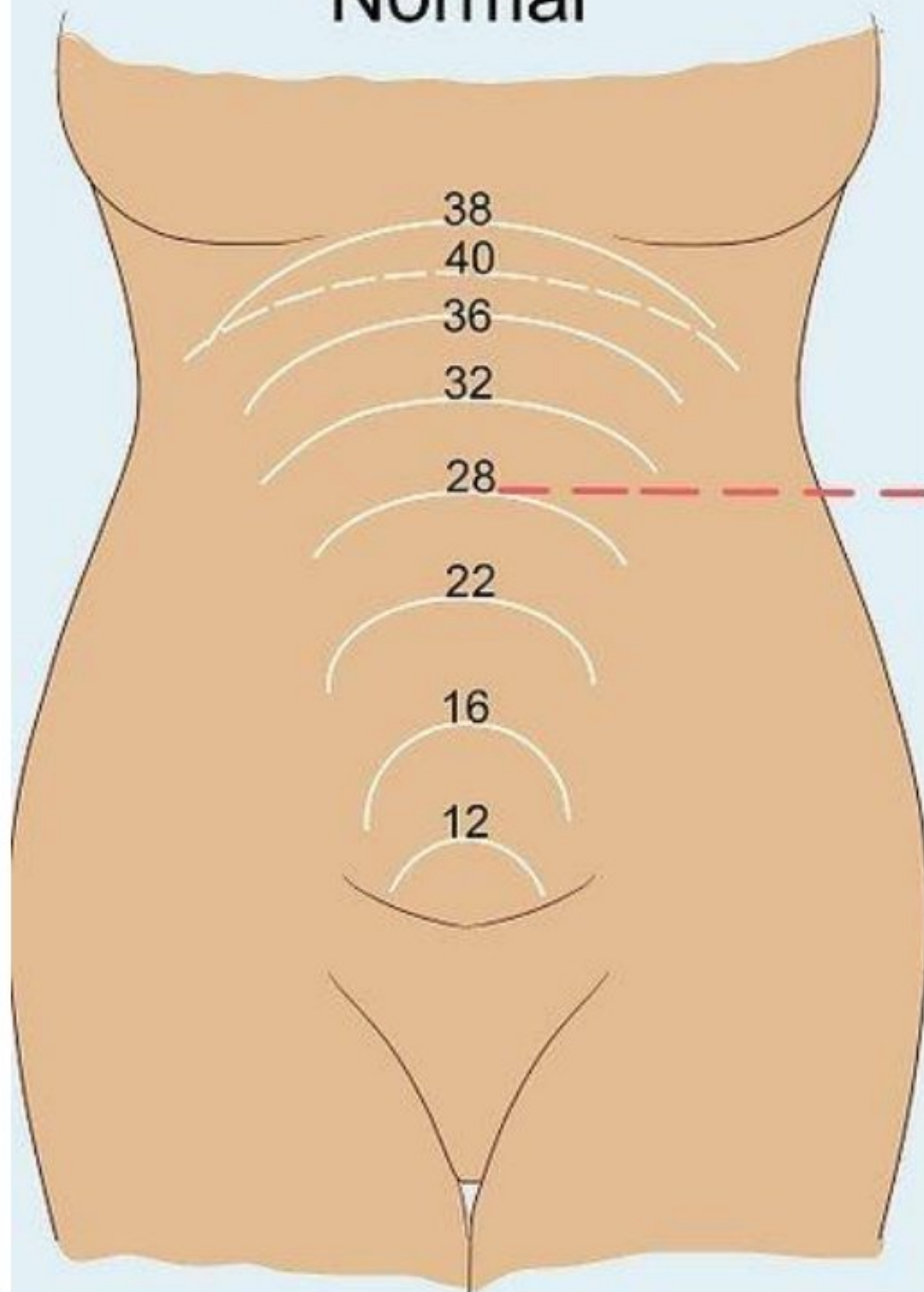
-At 12th week: just palpable above the symphysis pubis

- At 16th week: lower one-third of the distance between the symphysis pubis and umbilicus
- At 20th week: two-thirds of the distance between the symphysis pubis and umbilicus
- At 24th week: at the level of the umbilicus
- At 28th week: lower one-third of the distance between the umbilicus and xiphisternum

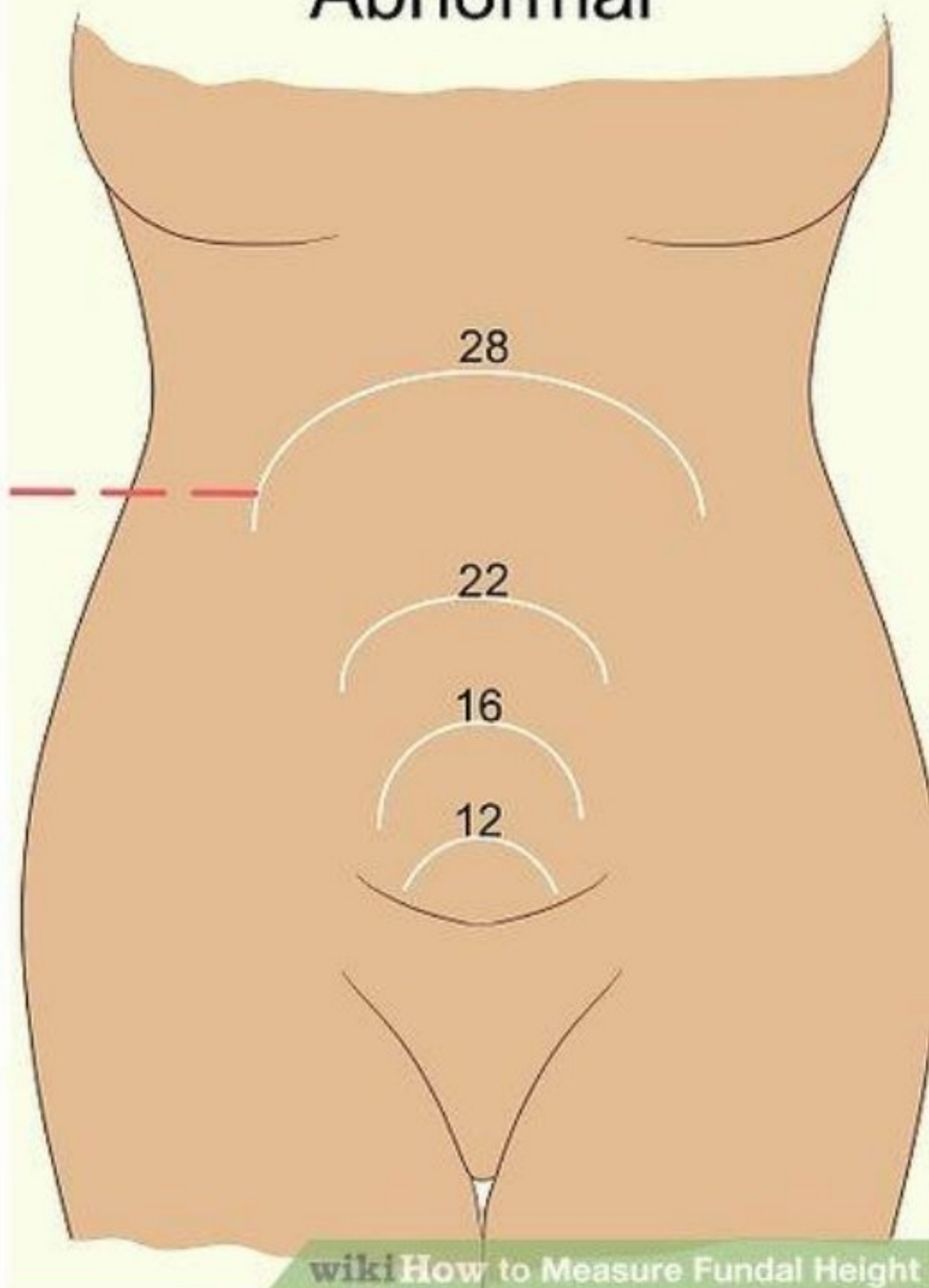
How to Determine Fetal Lie and Presentation

- At 32nd week: two-thirds of the distance between the umbilicus and xiphisternum
- At 36th week: at the week of the xiphisternum
- At 40th week: sinks back to the level of the 32nd week, but the flanks are full, unlike that in the 32nd week

Normal



Abnormal



wikiHow to Measure Fundal Height

Abdominal grips.

The pelvic grips (four in number) are performed to determine the lie and the presenting part of the fetus

- **Fundal palpation/fundal grip**

This palpation helps determine the lie and presentation of the fetus

Palpate the uterine fundus gently by laying both hands on the sides of the fundus in an attempt to determine which pole of the fetus (the breech or the head) is occupying the uterine fundus.

The head feels like a hard globular mass which is ballotable (moves between the fingertips of the two hands), whereas the breech is of a softer consistency and has an indefinite outline

In the case of a transverse lie, the fundal grip will be empty

- **Lateral palpation/lateral grip**

This palpation is used to locate the fetal back to determine the fetal lie

Place the hands on either side of the uterus at the level of the umbilicus and apply gentle pressure. The back of the fetus is felt like a continuous hard, flat surface on one side of the midline and the limbs are felt as irregular small knobs on the other side

In the case of a transverse lie, the back is felt transversely, i.e. stretching across both sides of the midline

- **First pelvic grip/superficial pelvic grip**

The third maneuver must be performed gently, or it will cause pain to the woman. Spread your right hand widely over the symphysis pubis, with the ulnar border of the hand touching the symphysis pubis. Try to approximate the finger and thumb, putting gentle but deep pressure over the lower part of the uterus. The presenting part can be felt between the fingers and the thumb. Determine whether it is the head or the breech (in the case of a longitudinal lie)

The mobility of the presenting part can also be determined by gripping the presenting part and trying to move it. If it cannot be moved, it indicates that the presenting part is “engaged”. The fetal head is said to be engaged if the widest diameter of the fetal head has passed through the brim of the pelvis, or only two finger-breadths are felt above the pelvic brim

In the case of a transverse lie, the third grip will be empty

- **Second pelvic grip/deep pelvic grip**

To perform this grip, you must face the foot end of the mother. Keep both the palms of your hands on the sides of the uterus, with the fingers held close together, pointing downward and inward, and palpate to recognize the presenting part

If the presenting part is the head (felt like a firm, round mass, which is ballotable, unless engaged), this maneuver, in experienced hands, will also be able to tell us about its flexion

If the woman cannot relax her muscles, tell her to flex her legs slightly and to breathe deeply.

Palpate in between the deep breaths

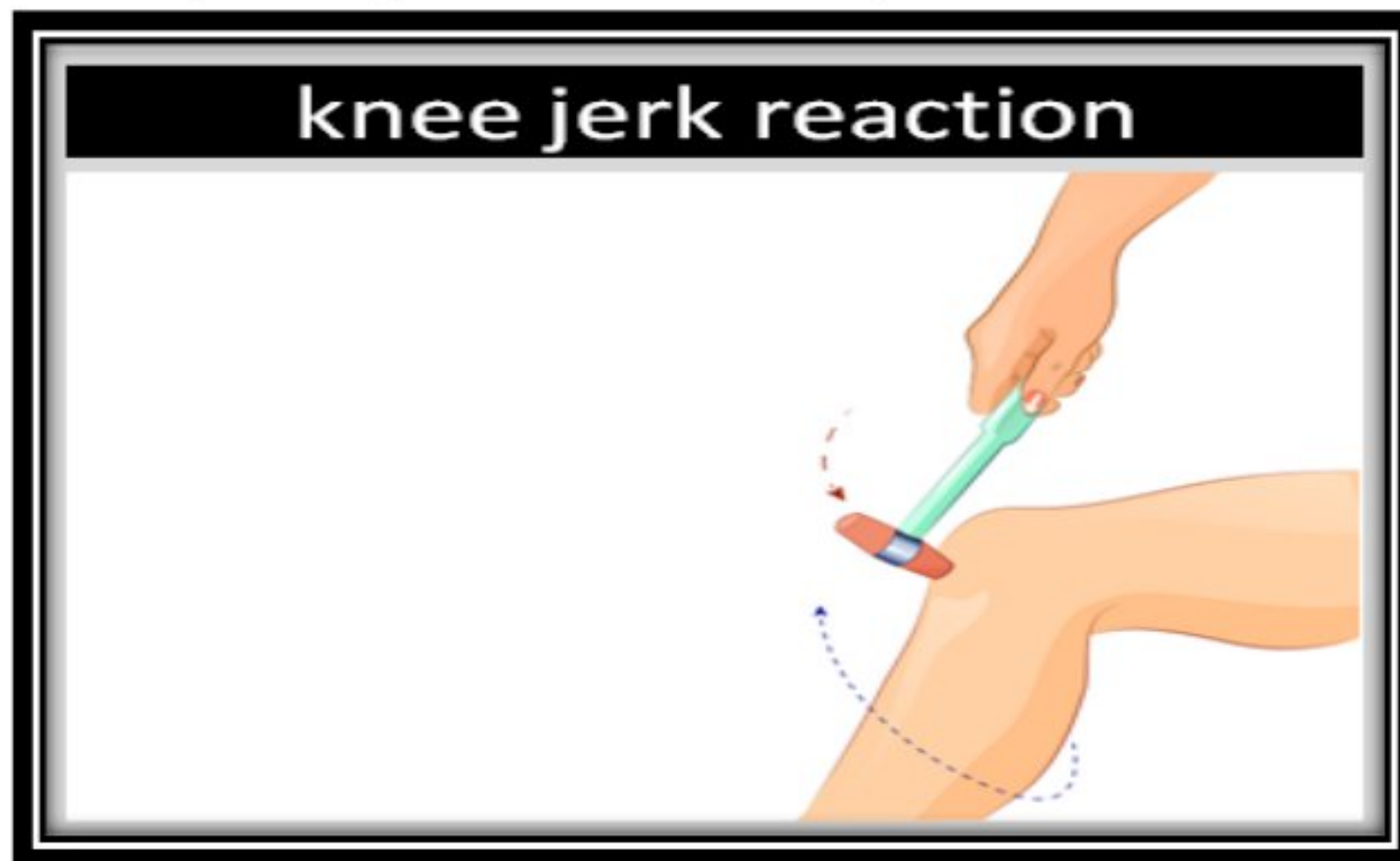
Routes to give magnesium sulfate.

NO	Steps
Administering Loading Dose (IV+ IM) of Magnesium Sulfate	
1	Washes hands thoroughly with soap and water and air dry. Puts clean examination gloves on both hands.
2	Prepares magnesium sulfate 20% solution, 4 g (Take one 20ml sterile syringe, draw 4 ampoules of Mg So ₄ (8ml=4g) into the syringe, add 12 ml of distilled water /normal saline for injection to make it 20%).
3	Carefully cleans the injection site with an alcohol wipe.
4	Gives magnesium sulfate 20% solution, 4 g by IV injection SLOWLY over 5 minutes.
5	Disposes of used needle and syringe in a sharps disposal box.
Administering IM loading Dose of Magnesium Sulfate	
6	Prepares 2 syringes (10ml syringe with 22 gauze needle) with 5 g of 50% magnesium sulfate solution with 1 mL of 2% Lignocaine in the same syringe.
7	Carefully cleans the injection site with an alcohol wipe.
8	Gives 5 g by DEEP IM injection in one buttock.
9	Disposes of used needle and syringe in a sharps disposal box.
10	Carefully cleans the injection site in the other buttock with an alcohol wipe.
11	Gives 5 g by DEEP IM injection into the other buttock.
12	Disposes of used needle and syringe in a sharps disposal box.
13	Disposes of gloves.
14	Washes hands thoroughly with soap and water then air dry.
15	Records drug administration and findings on the woman's record.
Administering IV Dose of Magnesium Sulfate for recurrent fits / convulsions	
16	Washes hands thoroughly with soap and water and air dry. Puts clean exam gloves on both hands.
17	Prepares syringe with 2 g magnesium sulfate (50% solution) Take one 10ml sterile syringe, draw 2 ampoules of Mg So ₄ 50%(4ml=2g) into the syringe add 6 ml of distilled water /normal saline.
18	Carefully cleans the injection site with an alcohol wipe.
19	Gives magnesium sulfate 20% solution, 2 g by IV injection SLOWLY over 5 minutes.
20	Disposes of used needle and syringe in a sharps disposal box.
21	Disposes of gloves.
22	Washes hands thoroughly with soap and water and dries with a clean, dry cloth or air dry.
Maintenance dose of MgSo₄	
23	Washes hands thoroughly with soap and water and air dry. Puts clean exam gloves on both hands.
24	Prepares 1 syringe(10ml syringe with 22 gauze needle) with 5 g of 50% magnesium sulfate solution with 1 mL of 2% Lignocaine in the same syringe
25	Carefully cleans the injection site with an alcohol wipe.

26	Gives 5 g by DEEP IM injection every 4 hourly in alternate buttock.
27	Maintenance dose of MgSO ₄ to be continued till 24 hours after delivery or the last convulsion whichever is later.
28	Disposes of used needle and syringe in a sharps disposal box.
29	Disposes of gloves.
30	Washes hands thoroughly with soap and water and dries with a clean, dry cloth or air dry.
31	Records drug administration and findings on the woman's record.

Notes

- DO NOT give next dose of Mg SO₄ if absent knee jerk or urinary output less than 100ml/4 hours or respiratory rate less than 16/min.



- Signs of reaction: After receiving the injection, the woman may have flushing, may feel thirsty, get a headache, feel nauseous or even vomit.
Calcium Gluconate, 10%, 10ml as an antidote of M

Perennial care

Introduction

Perennial care involves washing the genital and rectal areas of the body. It should be done at least once a day during bed bath, shower, or tub bath

Objectives:

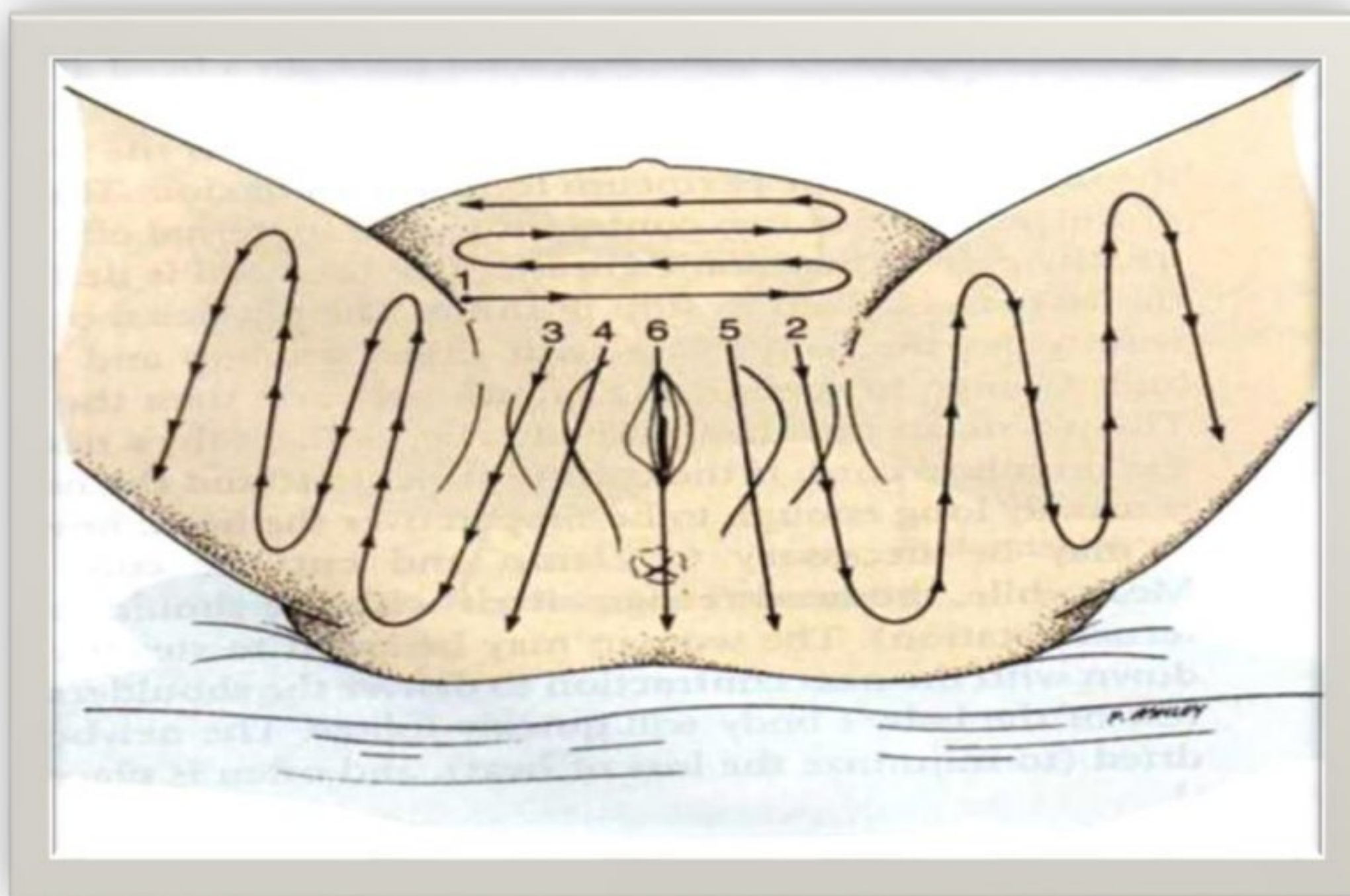
- To maintain the perennial area clean and dry
- To prevent postpartum complications as puerperal sepsis
- To assess integrity of perennial area

Definition:

perennial care involves washing the external genitalia and surrounding with soap and water or with water alone or in combination with any commercially prepared per-wash

Principle:

- Clean the perineum from the cleanest to less clean area.
- Patient who require special attention to perennial area.
 - Patient who are unable to do self-care.
 - Patient with genitor-urinary tract infection.
 - Patient with incontinence of urine and stool.
 - Patient with indwelling catheters.
 - Postpartum patients.
 - Patients after surgery on the genitor-urinary system.
 - Patients with injury, ulcer or surgery on perennial area



Procedure

Procedure	Rationale
1. The following equipment is needed for this skill: a. towels b. gloves c. bath basin & warm water d. soap e. washcloth f. protective pads/linen g. cotton balls/gauze.	Water is to be warm to allow comfort. Too hot may damage skin and cold may be startling and uncomfortable. Some facilities may have bath blankets and prepared cleansing tissues or towels.
2. Prepare the patient by folding the top bed linen to the foot of the bed and fold up the gown to expose the perineal/genital area.	Assessment cannot be completed unless the genital area is visible.
3. Place a bath towel under the patient's hips.	Prevents the bed from becoming soiled.
4. Position and drape the patient.	Draping the patient minimizes exposure and provides additional warmth for the patient.
5. Have the patient lie on their back with knees flexed and spread apart.	Allows adequate visibility and access and flexing the knees decreases the stress on the lower back.
6. Drape the legs by covering the torso with an additional sheet or blanket and tuck the corners under to inner sides of the legs. Allow some of the sheet to cover the pubic area until cleaning commences.	A separate top sheet may be used to provide additional warmth and comfort to the patient. Decreasing the upper thigh exposure and allowing a loose drape to cover the pubic area decreases the sense of exposure on the part of the patient.
7. Clean the labia majora, then spread the labia to wash the folds between the labia majora and the labia minora.	Secretions in this area tend to collect and facilitate bacterial growth.
8. Clean in direction from perineum to rectum.	Wiping in this manner prevents cross contamination of areas.
9. Using separate quarters of the washcloth for each stroke, wipe from the pubis to the rectum.	Using clean portions of the cloth discourages transmission of microorganisms from one area to another. Wiping from an area of least contamination to one of more contamination prevents spreading microorganisms.

Procedure	Rationale
10. For menstruating patients or those with indwelling catheters, disposable cloths, gauze, or cotton balls may be useful. Use a clean area for each stroke.	Additional soiled areas may require the use of additional clean supplies that promote easy disposal of soiled surfaces.
11. With a catheter, wipe in the direction away from entry site.	Wiping away from the body portal prevents microorganisms from entering into the body cavity.
12. Rinse well. May use a peri-wash bottle or container to flow warm water over the perineal area. Having the patient on a bedpan allows collection of the draining fluid.	Soap residue may act as an irritant and harbor microorganisms. Allowing a free flow of fluid is soothing to the patient and increases the effectiveness of removing residue from the area.
13. Dry thoroughly, paying attention to the folds between the labia.	Moisture supports and promotes the growth of many microorganisms.
14. Assess the area for evidence of redness, irritation, and drainage, especially between the labia in females and around indwelling catheters.	Indicates skin breakdown.
15. Reposition the patient on the side facing away and clean between the buttocks. May use tissue first if heavily soiled.	Provides visual inspection of the area. Tissue is more readily disposable and reserves the washcloth for the final cleaning.
16. Dry the area well.	Moisture supports bacterial growth.
17. Assess for redness, excoriation, tenderness, discharge, or drainage.	Indicates skin breakdown.
18. Document condition of skin.	All data must be entered in patient's record.
19. Report any abnormal data to the appropriate personnel.	Any abnormal finding must have a corresponding nursing action.

Procedure: uterine message

Objectives:

The student will be able to:

1. Define uterine message & it's purpose
2. Discuss the technique of uterus message
3. Demonstrate uterine message

Definition:

It is the manipulation of the postpartum uterus through the abdominal wall to prevent the risk of postpartum hemorrhage due to uterine atony.

Purpose:

- To help encourage the uterus continue to contract.
- To prevent postpartum hemorrhage.

Equipment:

- Gloves

Checklist: Uterine message

	Procedure	Rationale
1	Explain the procedure to the client	To alleviate anxiety
2	Keep the patient privacy.	To preserve the patient integrity
3	Ask the patient to void (unless bleeding is extensive and more rapid action seems necessary).	To prevents the uterus from deviating
4	Place the patient supine with her knees flexed and feet together.	To start the procedure
5	Put on gloves.	To prevent cross infection
6	Place one hand on the abdomen just above the symphysis pubis. Place the other hand around the top of the fundus.	Putting a hand on the symphysis pubis prevents the uterus from flipping.



- 7 Rotate the upper hand to massage the uterus until it is firm, being careful not to over massage the uterus.



over massaging the uterus may hurt the patient.

- 8 When the uterus is firm, gently press the fundus between the hands using slight downward pressure against the lower hand.

To end the fundus massage

- 9 Remove and observe the woman's perineum for the passage of clots and the amount of bleeding.

To assess any abnormalities

- 10 Massage the uterus again to be certain it remains firm, cleanse the perineum and apply perineal pad.

For sanitation purposes

- 11 Document the results of the procedure. Notify the physician if fundus does not remain firm or if bleeding continues.

To provide right intervention

- 12 Ability to answer questions

To be competent as a health care provider.

Procedure: breast examination

Objectives:

The student will be able to:

1. Define breast examination.
2. Discuss the techniques in doing physical breast examination.
3. Apply breast examination.

Definition:

Breast Self-examination is a common screening method which is done by the patient herself as an effort to detect the presence of breast cancer in its earliest stages by physically examining both the breasts for the possibility of any lumps, distortions, or swelling.

Purpose:

- Determine changes that may indicate infection or cancer (such as lumps or spots that feel different).

Equipment:

- Personal protective equipment

Checklist: Breast Examination

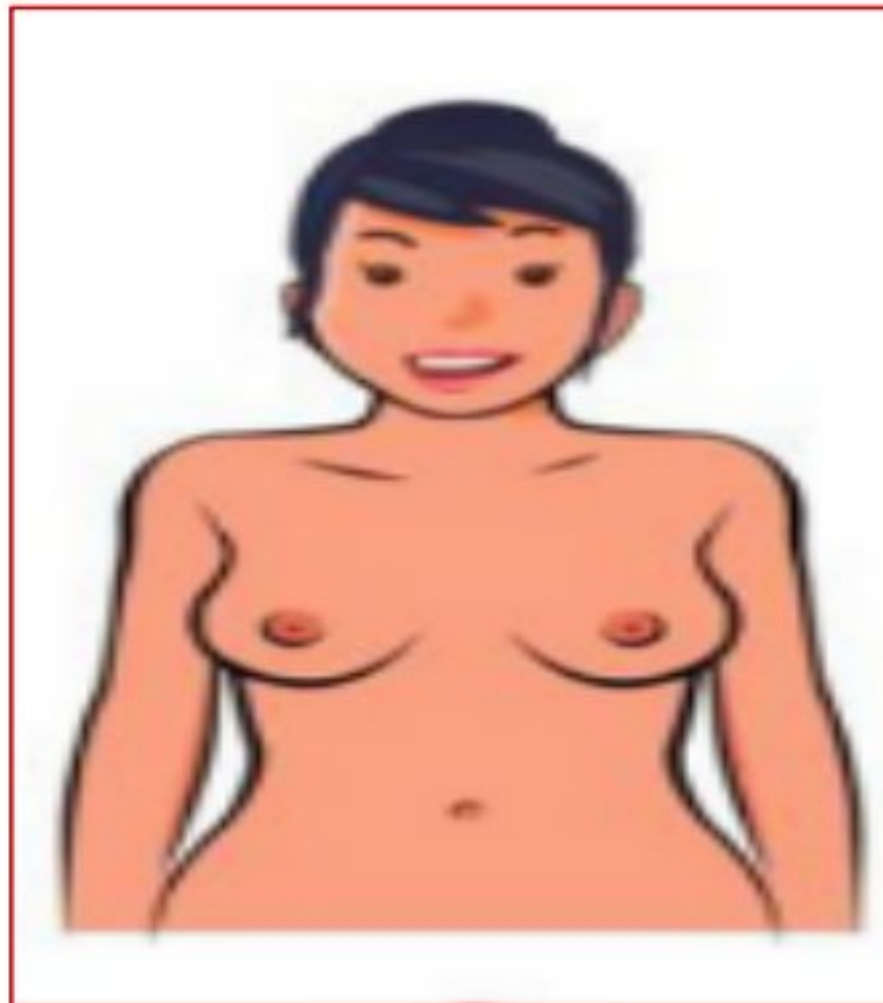
	Introduction
1	Wash your hands and don PPE if appropriate
2	Introduce yourself to the patient including your name and role
3	Confirm the patient's name and date of birth
4	Briefly explain what the examination will involve using patient-friendly language
5	Explain the need for a chaperone and gain consent to proceed with the examination
6	Position the patient sitting upright on the side of the bed
7	Adequately expose the patient's breasts
8	Ask if the patient has any pain before proceeding

Inspection

- Obvious masses
- Scars
- Skin changes or ulceration
- Asymmetry
- Nipple changes or discharge

9

Inspect the breasts with the patient's arms by their sides



10

Inspect the breasts with the patient's hands pushing into their hips

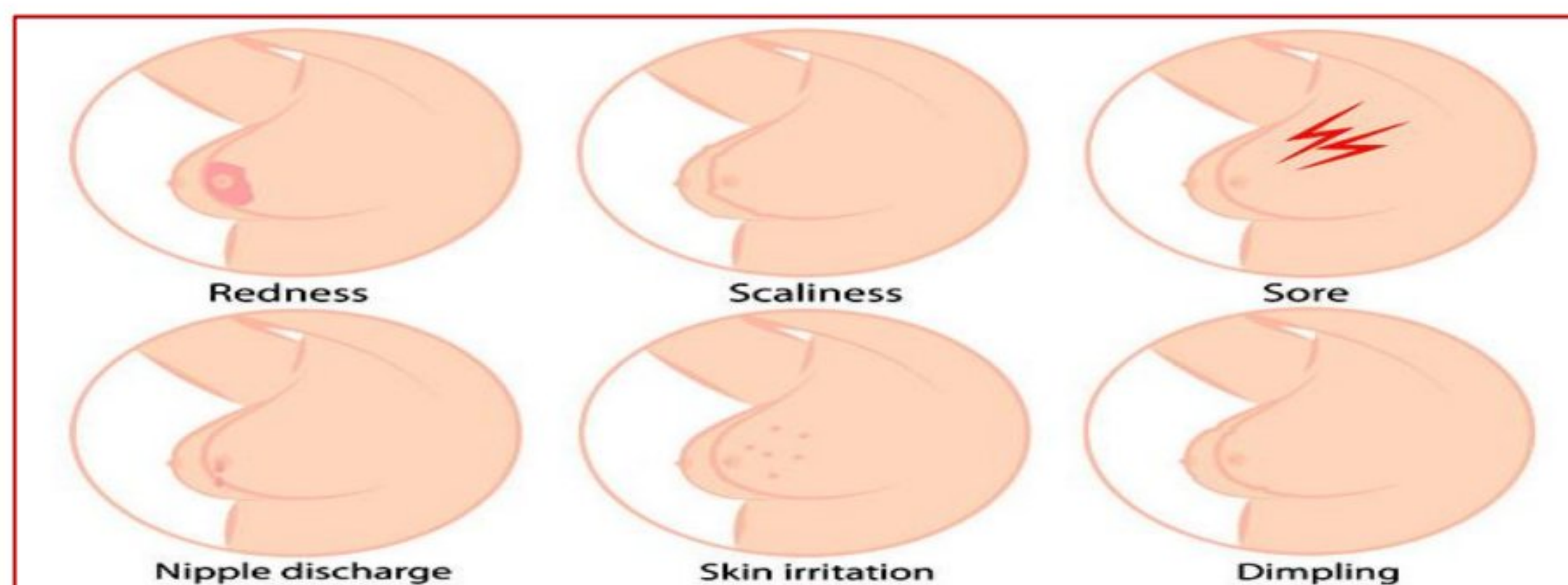


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a. Inspect the breasts with the patient's arms above their head whilst leaning forward.



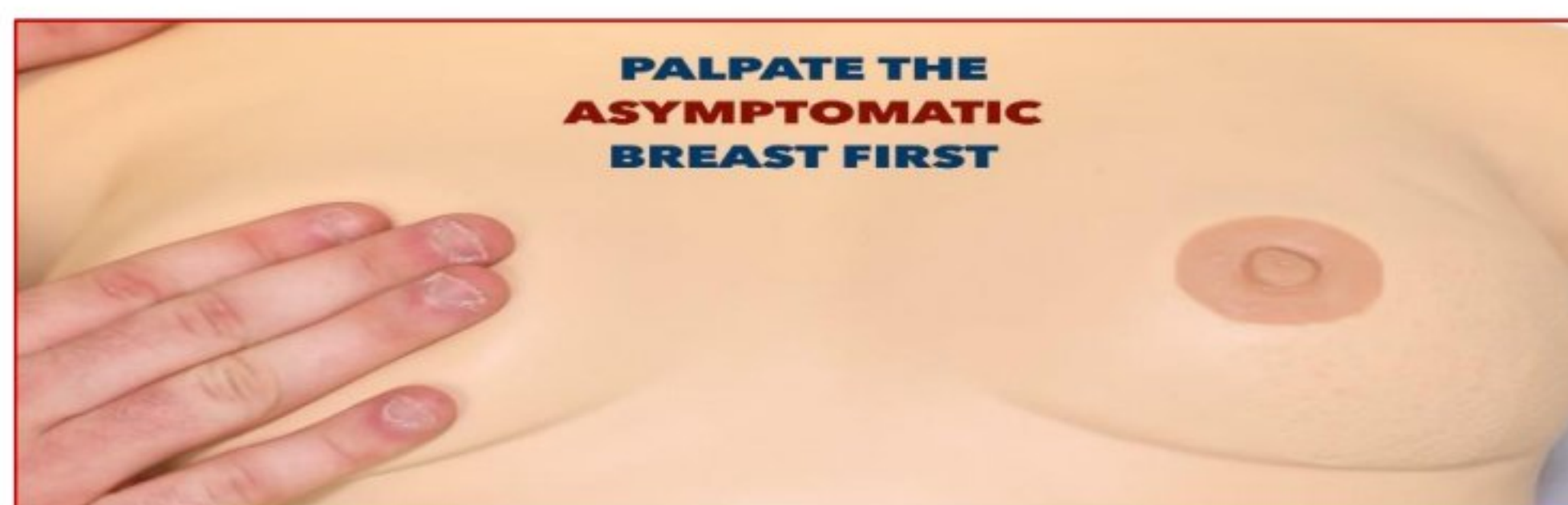
a. Elevate the breast and inspect for hidden pathology



Palpation: There are different ways to move across the breast:



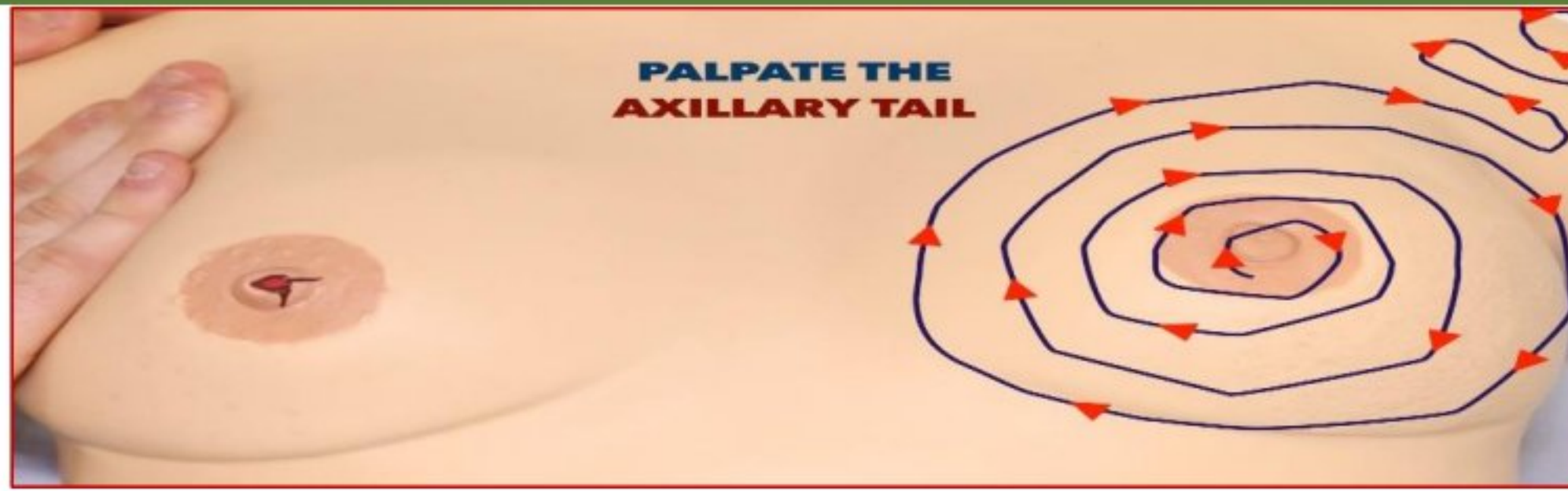
Palpate each breast systematically



12

13

Palpate each axillary tail



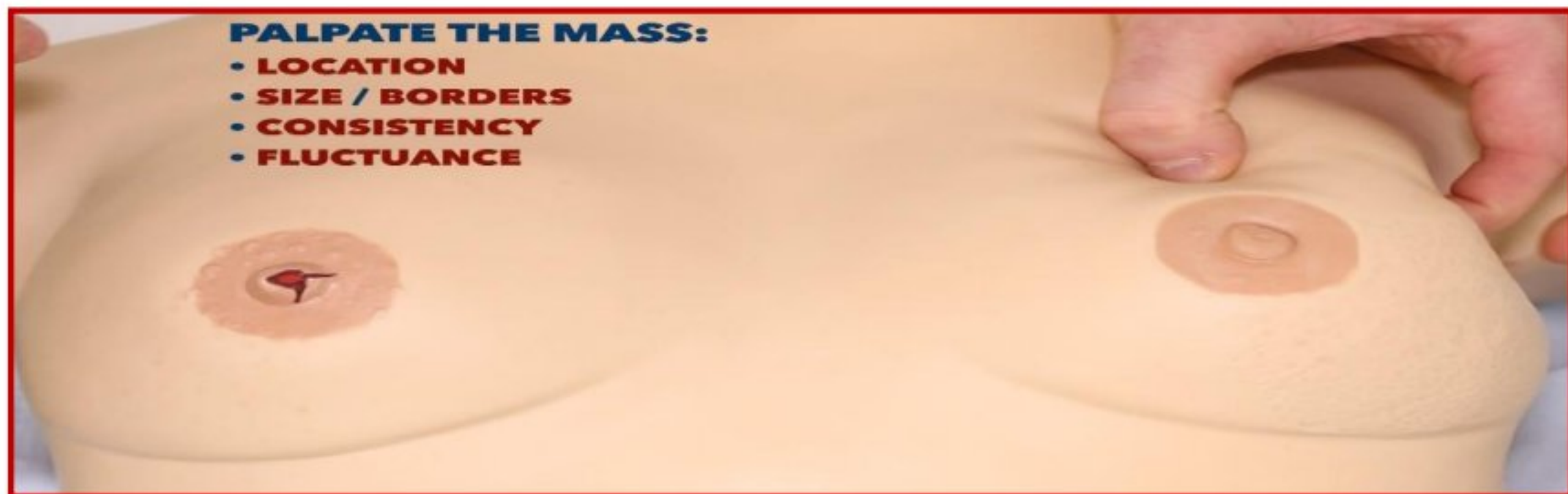
14

Palpate the nipple-areolar complex



15

Palpate for any mass



Palpate the axillary lymph nodes



Palpate the cervical, supraclavicular, infraclavicular and parasternal lymph nodes

To complete the examination

- | | |
|----|--|
| 17 | Explain to the patient that the examination is now finished & thank the patient for their time |
| 18 | Dispose of PPE appropriately and wash your hands |
| 19 | Summarise your findings |
| 20 | Suggest further assessments and investigations (e.g., biopsy, mammography, ultrasound) |

Breast self-examination

By Looking

Stand before a mirror and look at both breasts and nipples.



Once with both arms beside your body



Once with both hands on your hips



Once with both arms raised and arms behind your head

By Touching

Raise your left arm. Use the pads of your 3 or 4 fingers of the right hand to examine your left breast. Repeat on the right side.



Once in lines up and down your breast from outside to inside



Once in circles around the breast making smaller circles towards the nipples



Once in wedges starting from outside towards nipples and back again

While Lying Down

Lie on your back, left arm over your head and a pillow under your left shoulder. Examine your left breast with your right hand as described before. Repeat for the right side.

This position flattens the breast and makes it easier to examine.



Lymph Nodes

Place your left hand on your waist.

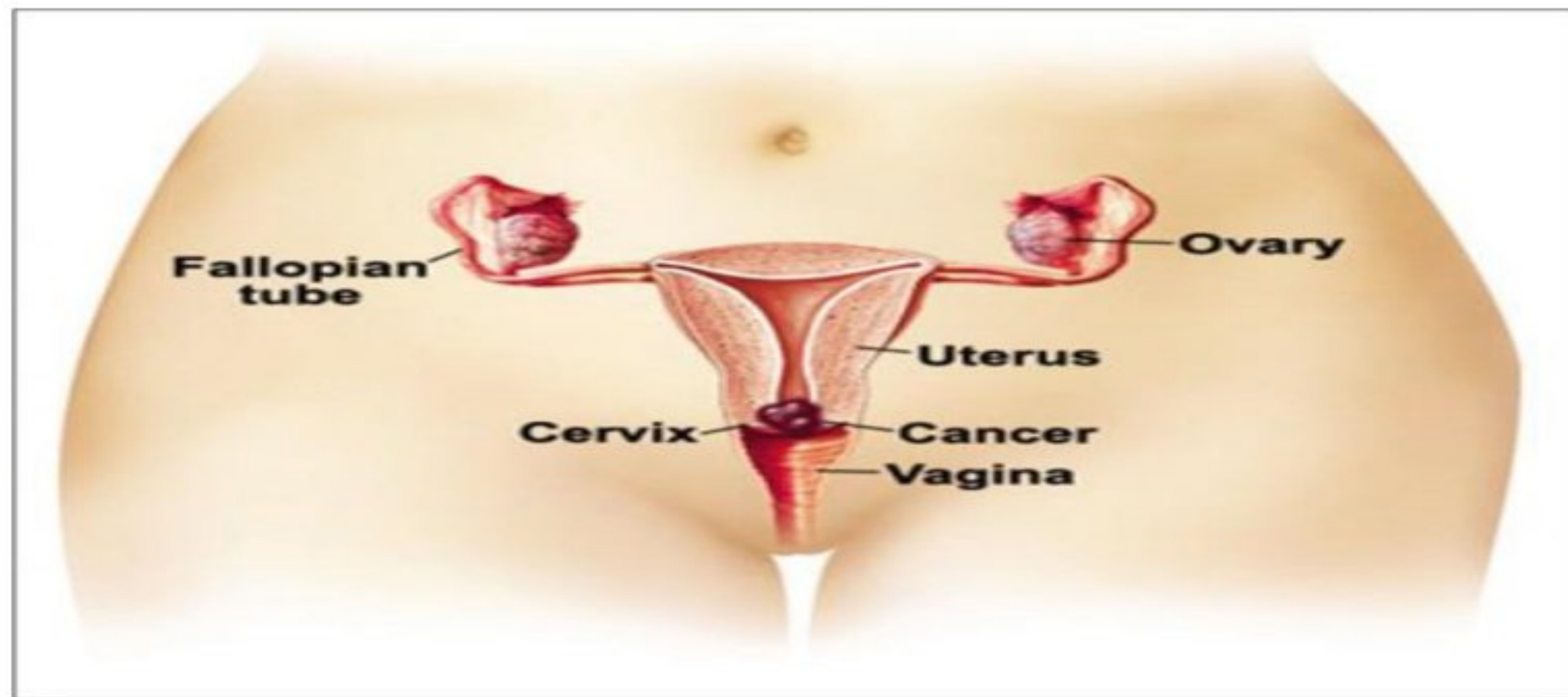
With your right hand try to find enlarged lymph nodes in the left underarm area and around your left collar bone. Repeat on the right side.



An enlarged lymph node feels like a corn kernel or a bean

Pap smear

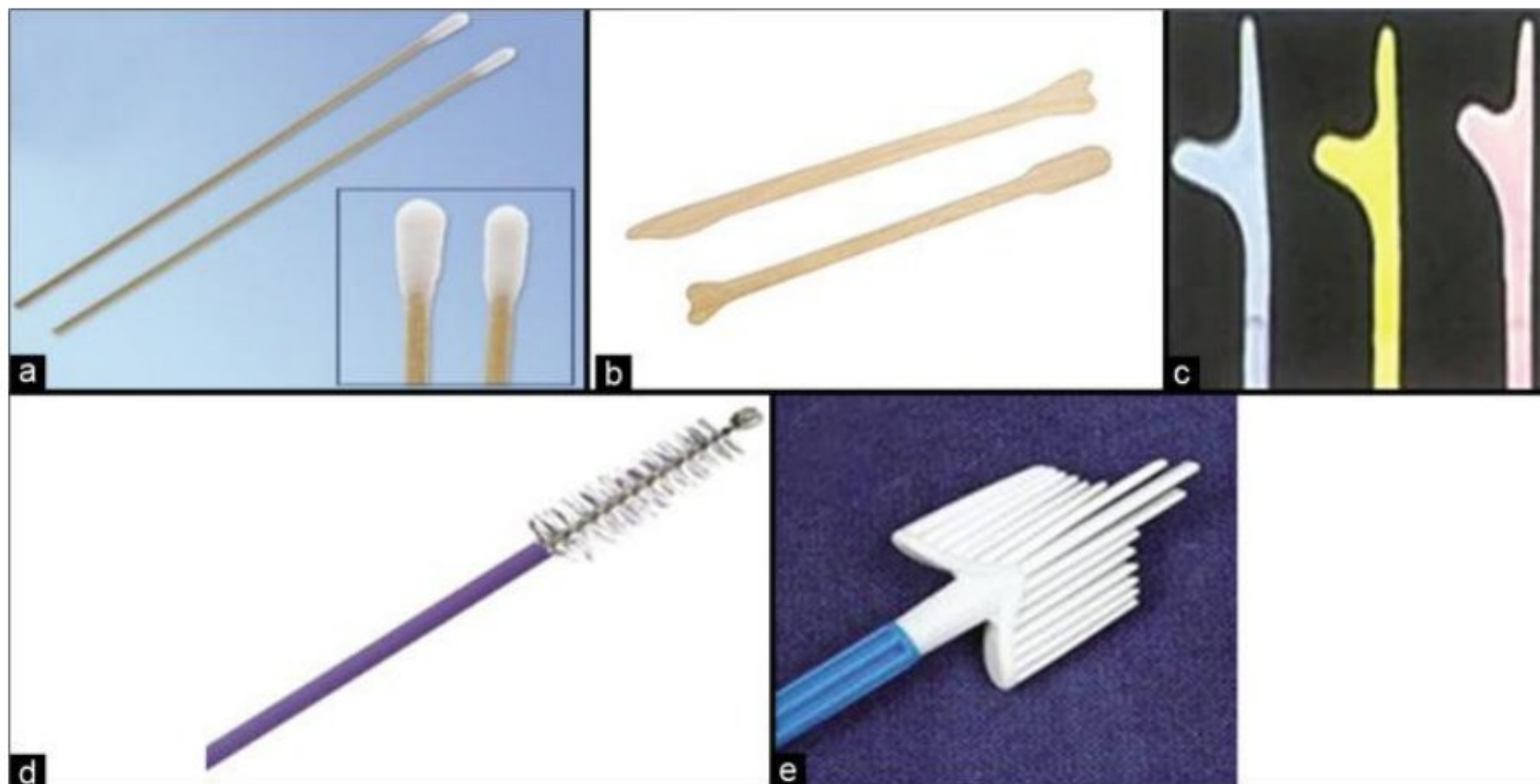
Definition: A Pap smear, also known as Papanicolaou smear, is a microscopic examination of cells scraped from the cervix and is used to detect cancerous or pre-cancerous conditions of the cervix or other medical conditions.



Anatomy of female genital tract and the location of cervical cancer

Basic equipment required for Pap smear collection:

1. Sterilized dry bivalve Cusco speculum
2. Clean glass slides labeled with a diamond pencil
3. Sterilized cotton tipped applicator/sticks/Ayres spatula or a Cervex brush



1. A wide mouth specimen bottle with fixative (absolute alcohol) or commercial instant spray fixative
2. Normal saline

3. Glass marking diamond pencil
4. Hanging drop slide and 22×22 mm glass coverslips to check for *Trichomonas vaginalis*
5. Curved metal cannula or a disposable plastic endometrial aspirator (in case of endometrial cytology).



Instruct the women

1. To come during the postmenstrual period or best between the 14th and 28th day of cycle (premenstrual period)
2. Not to come when there is active bleeding. First 3–4 days of menstruation are best avoided although not a contraindication
3. To avoid intercourse on the previous day, avoid douching with any antiseptic solution, not to insert any vaginal tablets as a part of ongoing treatment.
4. Not to come if a per vaginal examination is done in the past 24 h, or a cervical biopsy or cryotherapy or any other procedure has been done in the past 6 weeks.

Technique of collecting a pap smear:

Procedure	Rational
1-Explain the procedure to the patient and take a verbal or informed consent.	
2-The patient is instructed to sleep in a lithotomy position.	
3-A Sim's speculum or a bivalve Cusco's speculum is inserted in the vagina	-The lubricant jelly can cause artifacts in the smear

-Do not use lubricated Speculum.	
4-The cervix is exposed with gentle maneuvering of the speculum.	- Use of Cusco speculum makes one “hands free.”
5-The long end of the tip of any of the spatulas mentioned above or the tip of the brush is put in the endocervical canal and fixed gently but firmly and then rotated clockwise through 360° 2–3 times and 5 times if the Cervex brush of Surepath is used.	The method of Pap smear collection. (a) Long end of the Szalay's plastic spatula is seen fixed firmly in the endocervical canal and rotated clockwise through 360°, (b) The long end collects cells from the squamocolumnar junction even if it is higher up in the canal. (c) Soft bristles of Cervex brush of SurePath fit snugly on the contours of the cervix and when it is rotated 5 times through 360°, it collects cells from the complete transformation zone and also from the endocervical mucosa lining the canal.
6-The material is spread in a circular or linear fashion in the central 2/3rd area of the glass slide leaving the edges of the glass slide free of material.	A and b: Material is spread in a circular or linear fashion on a glass slide for preparing conventional smears.
7- For Liquid based Cytology (LBC) preparation, the Cervix brush is broken away or detached from the stick and tip of the Cervix brush is dropped in the LBC vial containing fixative	A. LBC vial. b.Head of Cervix brush being detached. c. LBC vial with brush dropped into the fixative.
8-The conventional smear is immediately fixed in absolute alcohol.	- Delay in fixation causes air drying artifacts and there is clouding of morphological details.
9-For spray fixing of the slide, spray should be kept at an angle of 45° and at a distance of 6 inches from the slide.	

Post-procedure: 10- Put the women in comfortable position. 11- Return equipment. 12- Hand wash. 13- Documentation.	